JAVA PROGRAMMING PRACTICES

Roman Numeral Conversion Using a Package

Basic Information

Week	Week 9
Source file	RomanDemo.java
Concepts	Packages, imports, classes, methods, string handling.

Objective / Aim

To implement roman numeral conversion using a package using the concepts listed in the R23 Java Programming Practices syllabus.

System Requirements

JDK (Java Development Kit) and a text editor or Java IDE.

Algorithm / Logic Description

1. Create number.Roman and implement romanToInteger().
2. Import the class in RomanDemo.
3. Read a Roman numeral and display its integer value.

UML / Program Structure

```
number.Roman
+ romanToInteger(text)
RomanDemo imports number.Roman
```

Source Code

```
// File: number/Roman.java
package number;
public class Roman
{
    public int romanToInteger(String text)
    {
        int total = 0, previous = 0;
        for (int i = text.length() - 1; i >= 0; i--)
        {
            int value = valueOf(text.charAt(i));
            if (value < previous) total -= value;
            else total += value;
            previous = value;
        }
        return total;
    }
    private int valueOf(char ch)
    {
        switch (ch)
        {
            case 'I': return 1;
            case 'V': return 5;
            case 'X': return 10;
            case 'L': return 50;
            case 'C': return 100;
            case 'D': return 500;
            case 'M': return 1000;
            default: return 0;
        }
    }
}

// File: RomanDemo.java
import java.util.Scanner;
import number.Roman;
public class RomanDemo
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter Roman numeral (uppercase): ");
        String text = sc.nextLine().trim();
```

Source Code (continued)

```
        Roman roman = new Roman();
        System.out.println("Integer value: " + roman.romanToInteger(text));
        sc.close();
    }
}
```

Compilation and Execution

```
javac number/Roman.java RomanDemo.java
java RomanDemo
```

Output Screen

```
Enter Roman numeral (uppercase): Integer value: 58
```

Result

The program demonstrates the listed syllabus concepts. The sample output is based on a compiled execution.